LSE Department of Economics

MSc Extended Essay Cover Sheet

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COURSE CODE	E	C	4	2	3

TITLE

Is Artificial Intelligence Coming for Our Jobs? – Country-Industry Effects of AI on Labour Markets

WORD COUNT	5	8	9	6

Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text).*

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Is Artificial Intelligence Coming For Our Jobs? - Country-Industry Effects of AI on Labour Markets

Artificial Intelligence impresses with its capabilities and causes widespread fear about its effects on jobs and wages. Although there are multiple studies on the expected impact as well as the firm and individual-level effects of AI, there is little evidence on industry effects, especially outside the US. Using data on investments in AI startups from OECD.AI, I estimate the effects of AI on growth in employment, wages, capital stock and labour productivity in a country-industry cell. Trying to mitigate endogeneity issues, I propose two instrumental variables based on the interaction with the share of workers in ICT occupations. I find negative effects on employment and hours worked but positive ones for wages and labour productivity. Due to limitations of the research design and the data used, these results should be interpreted with caution.

I. Introduction

Do you think a human has written this abstract? I can assure you that I have written it and that I am, indeed, a human. Posing a question like this would have been considered extremely odd a few years ago but now it is becoming a sensible thing to ask, given that AI like ChatGPT can write academic abstracts (Gao et al., 2023) and the regulations applying

to this essay explicitly prohibit its use. Machines are increasingly entering domains of human activity which were considered impregnable before. For instance, AI algorithms can create works of art that are sold in auctions (Quackenbush, 2018) and win a photography award (Parshall, 2023). If even the arts and sciences have now become the domain of machines, what remains for us humans? Will artificial intelligence come for our jobs and render us redundant or is this hysteria?

To get a clearer picture about how much of a threat AI poses to employment, one can look at studies trying to predict the effect of AI. Frey and Osborne (2017), for instance, estimate that 47 percent of US employment is highly susceptible to computerisation. Given the labour market impacts of earlier automation technologies, like employment and wage polarisation for PCs (Autor and Dorn, 2013) or negative local employment and wage effects for industrial robots (Acemoglu and Restrepo, 2020), it is imperative to examine the effect AI has already had and whether something similar is to be expected as the technology matures.

Since the technology is still in its infancy, there is scant evidence. Felten, Raj and Seamans (2019) examine the exposure to AI at the occupation-state level and find a small positive impact on wages but none for employment. Using multiple measures, Fossen and Sorgner (2019) conclude that at the individual-level, AI advances are likely to ameliorate job stability and increase wage growth. Acemoglu et al. (2022), mainly analysing firm-level data, find that exposed firms reduce non-AI employment but they detect no industry-level impacts for the US. Given that this literature concentrates

on the US and has little to say about more aggregate effects, there is an important gap in the literature. This analysis tries to contribute to this literature by exploring country-industry effects of AI for the EU and the US. Furthermore, I exploit a different data set. While Acemoglu et al. (2022) use three different occupation-based measures for AI, which try to estimate some kind of susceptibility to AI, I use the OECD.AI data set on venture capital investments in AI start-ups (Tricot, 2021) as a measure of AI. I argue that most start-ups have a narrow range of business activities and sell their products and services for more specialised purposes, allowing me to contribute their impact to their industry of activity.

Using the change in AI VC investments per worker from 2012 to 2019 as a proxy for the AI intensity of a given country-industry cell, I estimate effects on a host of outcome variables, including the changes in log employed persons, hours worked, mean hourly wage, capital stock, and labour productivity. Trying to mitigate endogeneity concerns, I propose two instruments. The first one interacts the share of workers in ICT-related occupations in 2011 with the industry-level AI VC investments. I argue that it captures the effective investments of AI as skilled workers are needed to use the investments. The second one interacts the share of ICT workers with the change in the training compute of notable machine learning models, capturing the technological constraints of AI. I find negative significant effects for growth in employment and hours worked but positive significant effects for growth in hourly wages and labour productivity. Both instruments are not very strong but the results hold up to weak instrument robust inference. They

also fend off some threats to validity, although inference using the wild bootstrap only supports the findings for employment. There are multiple other potential issues like undetected endogeneity of the instruments and measurement problems of the OECD.AI data. Thus, I interpret these findings cautiously and conclude that more research is needed.

The remainder of this analysis is structured as follows. First, I present the theoretical framework of automation espoused by Acemoglu and Restrepo (2020), modify it and derive my empirical specification from it in section II. Then, I present the data as well as how they are transformed in section III. In section IV, I present my OLS and 2SLS findings as well as the robustness of my results. Moving to section V, I discuss the internal validity of my methodology and link my results to the existing literature. I conclude this work in section VI.

II. Theory and Specification

For a better and more structured understanding of the relationship of interest one wants to analyse, it is helpful to look at implications coming from economic theory. I use the task-based model delineated in Acemoglu and Restrepo (2020). Their model aims to look at the effect of robots on local labour market outcomes. However, instead of robots, one can more abstractly reason for any automation technology, as is done in the precursor version of the model outlined by Acemoglu and Restrepo (2018). As the authors try to identify local labour market effects with a shift-share design, the model has to be slightly modified to accommodate the different data

structure at hand. In this modified model, every country has a CES aggregate output, instead of every commuting zone. To keep the theoretical analysis simple, I abstract from trade between countries as it does not alter the core conclusion and since trade within a country can be assumed to be significantly more intensive as between countries. Every industry produces its product by using capital together with the quantity of tasks $x_{ci}(s)$, indexed by $s \in [0, 1]$. Central to this model is the fact that an automation technology, in this case AI, can be a substitute for labour for certain tasks $s \leq \theta_{ci}$ where $0 \leq \theta_{ci} \leq 1$ denotes the share of automatable tasks in each country-industry cell[1]. If γ_L is labour productivity and γ_H the productivity of AI then tasks are performed as follows:

$$x_{ci}(s) = \begin{cases} \gamma_M M_{ci}(s) + \gamma_L L_{ci}(s) & \text{if } s \leq \theta_{ci} \\ \gamma_L L_{ci}(s) & \text{if } s > \theta_{ci} \end{cases} \tag{1}$$

where M_{ci} denotes a measure of the usage of artificial intelligence and L_{ci} the number of workers. Assuming that AI is more productive than human workers for tasks that can be automated, all tasks that can be automated will be performed by AI and the rest by human workers. Assuming an initial state of automatable tasks $\theta_{ci} = 0$, meaning that at the start of the period of observation there was basically no automation via AI, one can calculate the effect of an increase in the ability to automate for employment L_{ci} and

[1] Acemoglu and Restrepo (2020) assume that technology only varies across industries. As I look at cross country differences, I make the additional assumption that it also differs across geographical units.

wages W_{ci}[2]:

$$d \ln L_{ci} = \zeta \frac{d\theta_{ci}}{1 - \theta_{ci}} \tag{2}$$

$$d \ln W_{ci} = \tau \frac{d\theta_{ci}}{1 - \theta_{ci}} \tag{3}$$

where ζ and τ are the effects of a change in automation technology on employment and wages. They are functions of the model parameters and the coefficient of interest for my empirical analysis. As $\theta_{ci} = 0$ in the beginning, the term in the denominator, which is the share of non-automatable tasks is around 1 and can be measured by $L_{ic,t=base}$, whereas the numerator can be measured by the change in the usage of AI, since only the usage of AI enters for the production of these tasks. Hence, for any outcome variable Y in this analysis, the above equation is estimated:

$$\Delta \ln Y_{ci} = \beta \frac{\Delta \text{AI Intensity}_{ci}}{L_{ci,t=2011}} + \mathbf{x}_{ci}^{\top}\gamma + \alpha_i + \delta_c + \epsilon_{ci} \tag{4}$$

where the change in AI intensity is measured by venture capital investments in AI, and $\mathbf{x}_{ci}$ denotes the various controls for demographic characteristics and potential confounders. I also include industry and country fixed effects.

[2]The derivation of these effects from the ones Acemoglu and Restrepo (2020) present can be found in the appendix.

III. Data

A. Outcome Variables

The main outcome variables in this analysis are the number of employed persons, the hours worked, and the mean hourly wage. All outcome variables come from the February 2023 release of the EUKLEMS - ITANProd data set (Bontadini et al., 2023). EU KLEMS includes all 27 EU countries as well as Japan, the UK and the US, although I drop the UK, Japan, and Malta from my main analysis due to the lack of data for the instrument. All the data are reported at the country-industry level. I then construct long differences of the main outcome variables, like Graetz and Michaels (2018), as I am interested in more long-term effects of AI:

$$\Delta \ln Y_{ci} = \ln Y_{ci,t=2019} - \ln Y_{ci,t=2012} \tag{5}$$

for any outcome variable Y observed in industry i and country c. Monetary outcome variables like wages or the capital stock are first converted into US Dollars using the annual exchange rates provided by the OECD (OECD, 2023). I also deflate them using the GDP deflator from the World Bank's World Development Indicators (World Bank, 2023) with base year 2015.

B. Measures of AI Intensity

To analyse the effect of AI on different labour market outcomes, one needs a measure of the amount of AI used. Due to the immaterial nature of artificial intelligence, this proves challenging. For my measure of AI, I resort to

venture capital investments in AI start-ups extracted from the OECD Artificial Intelligence Policy Observatory (Tricot, 2021). Artificial intelligence is here defined according to the OECD AI Principles stating that "*[an] AI system is a machine-based system that is capable of influencing the environment by producing an output (predictions, recommendations or decisions) for a given set of objectives. It uses machine and/or human-based data and inputs to (i) perceive real and/or virtual environments; (ii) abstract these perceptions into models through analysis in an automated manner (e.g. with machine learning), or manually; and (iii) use model inference to formulate options for outcomes. AI systems are designed to operate with varying levels of autonomy*" (OECD n.d. as cited in Tricot 2021, p.21). Venture capital investments in this data set comprise investments to private companies not quoted on the stock market and exclude secondary stock purchases, mergers or add-ons. Data about these investments come from the private firm Preqin and were processed by the Jožef Stefan Institute and subsequently analysed by the OECD (Tricot, 2021). Venture capital investments in AI start-ups are available from 2012-2021, although I only use the data until 2019 since I do not want to capture the influence of the pandemic. As the 20 industries reported in the OECD.AI data are not always congruent with the NACE Rev.2 industry classification (Eurostat, 2008) used in the EU KLEMS data, I have to merge some industries in both data sets to get industry classifications that are as congruent as possible with each other, resulting in 14 industries. The exact merging scheme can be seen in the appendix. Consistent with the reported theoretical foundation, I construct

the following measure of AI intensity:

$$\Delta\text{AI Intensity}_{ci} = \frac{\text{AI Investments}_{ci,t=2019} - \text{AI Investments}_{ci,t=2012}}{\text{Employed Persons}_{ci,t=2011}} \tag{6}$$

I deflate the AI investments and divide the difference by the employment counts in each country-industry cell in 2011, one year before the start of the period of observation. As there are serious endogeneity concerns of this measure, I try to instrument the change in AI VC investments. The first instrument is based on the interaction between the share of people employed in ICT related occupations in each country-industry cell in 2011 with the real AI VC investments for each industry, excluding the country being observed:

(7)

$$\text{Effective AI Investments}_{cit} = \frac{\text{Employment}^{ICT}_{ci,t=2011}}{\text{Employment}_{ci,t=2011}} \cdot \sum_{j \neq c} \text{AI Investments}_{jit}$$

Effective AI Investments are then transformed as in (6) to get the change in Effective AI Intensity. The data for the number of employed persons in ICT occupations come from two different sources. For the EU countries, data on the number of ICT professionals and ICT technicians for each industry come from the European Centre for the Development of Vocational Training (CEDEFOP, 2023). However, Malta does not report any persons employed in these occupations, so I omit it from my analysis. The data for the US is taken from the ACS 5-Year Estimates Public Use Microdata (United States Census Bureau, 2020). There is a slight mismatch between what the CEDEFOP calls ICT professionals and ICT technicians, as they use the ISCO-08

classification (ILO, 2012), and what the US data calls computer and mathematical occupations using the SOC (Executive Office of the President Office of Management and Budget, 2018). The occupations mostly overlap, however statisticians and data scientists, who are included in the US category, are not in the equivalent EU category. As equivalent data for the UK and Japan is missing at the desired level of aggregation, I omit them.

The second instrument uses the share of employed persons in ICT professions in 2011 as well but I instead interact it with the average training compute of notable ML systems from Sevilla et al. (2022):

(8)

$$\text{Effective AI Performance}_{cit} = \frac{\text{Employment}^{ICT}_{ci,t=2011}}{\text{Employment}_{ci,t=2011}} \cdot \overline{\text{Training Compute}_t}$$

The authors argue that there is a relationship between the capabilities of an AI algorithm and the so-called training compute in petaFLOPs[3], which allows to capture the technological advances of AI. Effective AI Performance is also transformed as in (6).

C. Controls

To account for potential confounders, I include some controls. Since investment in ICT capital is conceptually closely linked to investments in AI, I include a normalised long-differences measure for ICT investments as control. The data for ICT investments come from the EU KLEMS Capital

[3] "The training compute is the estimated computation to train the final version of an ML model. A FLOP is a floating-point operation, which is a standard measure of computation and equivalent to one addition, subtraction, multiplication, or division of two decimal numbers." (Our World in Data, 2023). A petaflop contains $10^{15} FLOPs$.

Accounts. Similarly, imports from China are demonstrated to have had significant effects on labour markets in the developed world, especially in the US (Autor, Dorn and Hanson, 2013). Using data from the UN Comtrade database (UN, 2023), I gather the value of imports from China for the countries of the analysis. Wherever I use imports from China one missing industry, real estate activities, is omitted. Again, I transform the data as the other independent variables. To account for compositional effects of the labour force, I include the share of hours worked by persons aged over 50, having attended university, and being female from EUKLEMS as controls. Since data on the employment shares of different demographic groups are missing for the US, I additionally gather the employment counts and hours worked of these demographic groups by industry from the ACS 5-Year Estimates Public Use Microdata (United States Census Bureau, 2020) and construct the equivalent shares.

IV. Results

A. Descriptive Statistics

Before presenting the main results of my empirical analysis, I first show some descriptive statistics to gain a better understanding of the evolution of venture capital investments in artificial intelligence.

Figure 1. Evolution of AI Intensity by Country

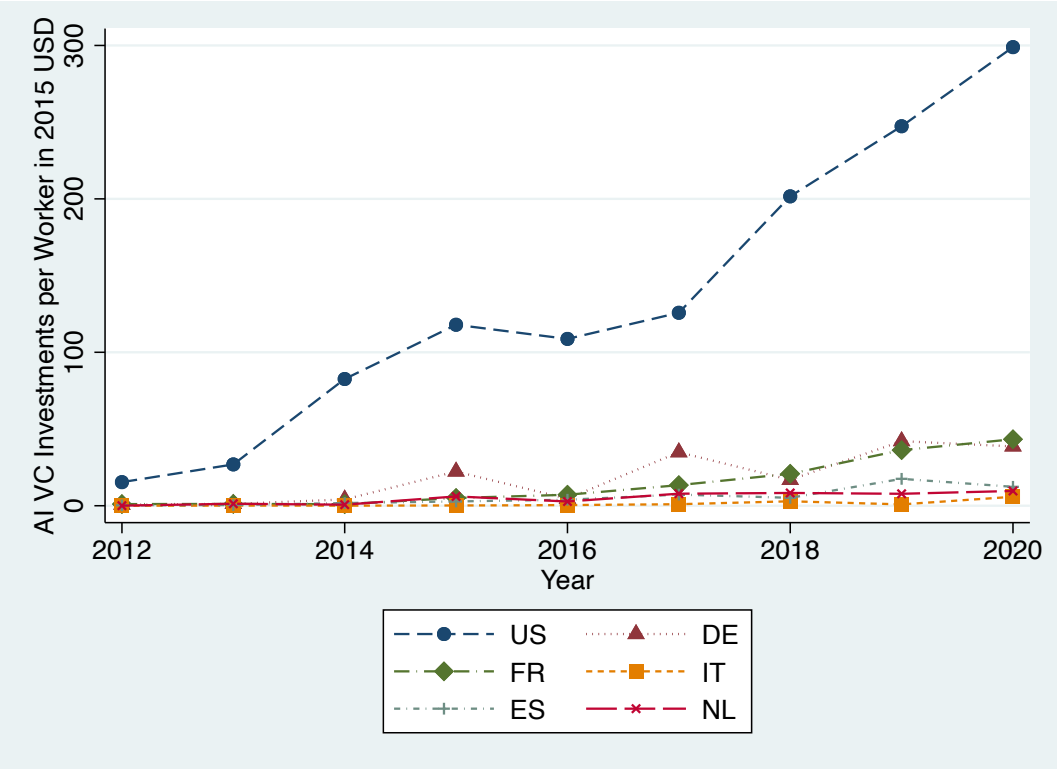
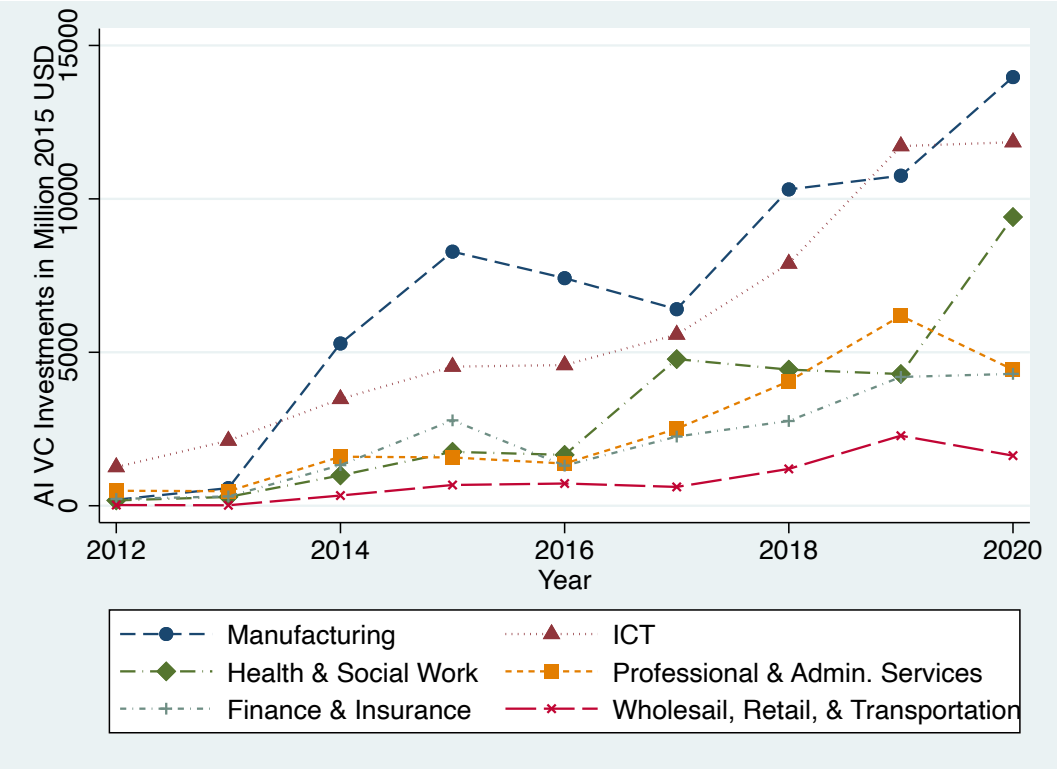


Figure 1 depicts the evolution of AI VC investments per worker in 2015 USD for the six biggest economies included in the analysis. There are significant differences in the intensity of AI VC investments. In every year, the US has by far the highest investments in AI per worker. The gap to the countries with the next highest investment density, France and Germany, only declines somewhat during the decade. Overall, AI investments per worker are on an upward trajectory for all countries, with the US even increasing the measure tenfold.

Figure 2. Evolution of AI VC Investments by Industry



Turning to Figure 2, one can see the total AI VC investments in million 2015 USD over the 27 countries of the analysis for the six industries with the highest AI investments. All these industries exhibit an enormous increase in total AI VC investments. For instance, the investments flowing to start-ups operating in the ICT industry almost increased tenfold in this period. One can see significant differences in levels and even trends. However, no one industry seems to dominate for all countries combined, unlike for the country-specific investments. These results suggest that the measure for intensity of AI investments introduced in section III.B is mainly concentrated

close to 0 with some large outliers.

B. OLS Results

Having gotten an intuitive sense for the AI data, it is now time for the main empirical analysis. I first run the specification presented in section II for employment, hours worked, and the log hourly wages. Table 1 summarises the results for both outcome variables. Panel A reports the effects of a change in AI intensity on the change in log employment, whereas Panel B reports these for the outcome variable change in log hours worked and Panel C summarises the results for the change in log real hourly wages. I weight by the initial shares of total country employment for each industry and weigh each country the same as is done by Graetz and Michaels (2018), who look at the effects of robots on productivity and labour markets using the EU KLEMS data as well. Standard errors are robust and two-way clustered at the country and industry level. Column 1 is the basic specification without any controls and fixed effects. For all three outcome variables, I report a positive correlation with a change in AI intensity which is however statistically insignificant. In column 2, I add country and industry fixed effects, as there is heterogeneity in levels and trends across these dimensions. For both employment and hours worked the correlation changes sign and becomes negative and statistically significantly different from zero.

Column 3 adds the demographic controls, whereas column 4 adds the change in ICT investment intensity as a control.

Table 1—OLS Results for Employment, Hours Worked and Average Wages

	(1)	(2)	(3)	(4)	(5)
Panel A: Change in Log Employment					
ΔAI Intensity	0.100	-0.057*	-0.020	-0.002	0.002
	(0.060)	(0.025)	(0.027)	(0.013)	(0.011)
Adjusted R^2	0.003	0.581	0.619	0.692	0.703
Panel B: Change in Log Hours Worked					
ΔAI Intensity	0.105	-0.071**	-0.039	-0.017	-0.013
	(0.059)	(0.023)	(0.028)	(0.014)	(0.013)
Adjusted R^2	0.004	0.618	0.645	0.679	0.697
Panel C: Change in Log Real Hourly Wages					
ΔAI Intensity	0.047	0.073	0.063	0.042	0.025
	(0.076)	(0.046)	(0.050)	(0.029)	(0.037)
Adjusted R^2	-0.002	0.741	0.745	0.770	0.782
Observations	378	378	378	297	238
FE		Yes	Yes	Yes	Yes
Demographics			Yes	Yes	Yes
ICT Investments				Yes	Yes
Chinese Imports					Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: ΔAI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included in designated specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares.

Since information and communication technologies like computers are closely linked to AI and as there is evidence that they might have an impact like employment polarisation through the automation of routine tasks (Autor and Dorn, 2013), it is imperative to include this potential confounder as a control. Similarly, column 5 includes the change in the intensity of Chinese imports as a control, given the literature on import competition and its effects on labour markets, especially for manufacturing industries (Autor, Dorn and Hanson, 2013). Including these controls has the effect that around a third of observations are lost due to missing values for these controls. Coefficients for all outcome variables decrease in magnitude and the one for the change in log employment even switches signs again in the final specification. Neither OLS coefficient is statistically significant anymore. Hence, the OLS results suggest no significant association with AI VC investments and the log change in these outcome variables.

C. IV Results

However, there are numerous threats to the internal validity of the OLS estimates. There could be other confounding variables, whose long differences vary across countries and industries, affecting employment or wages while being correlated with investments in AI. Examples would be labour market shocks increasing wages and hence increasing the benefits of automation. Another issue is reverse causality. AI VC investments might have an effect on employment and wages or the other way around, which would completely alter the conclusion drawn. If, hypothetically, AI affected

employment negatively this would indicate a displacement of workers due to automation and its general equilibrium effects. However, if a reduction in labour inputs were the cause of AI investments, AI would not displace workers but could be used to alleviate labour shortages. Both cases would result in negative OLS coefficients. Furthermore, there is the problem of measurement errors. Measurement errors could enter at multiple stages but mainly through missing and estimated AI investments in the OECD.AI dataset. If this measurement error is classical, it might result in an attenuation bias. However, the measurement issues in the AI data seem to be dependent on the country (Tricot, 2021), complicating the predicted effect on the coefficients. All these problems render a causal interpretation of the results in table 1 impossible.

To combat these issues, I implement an instrumental variable approach, aimed at isolating the exogenous variation in AI VC investments. I propose two different instruments. The first one is the change in Effective Intensity of AI as described in section II.B. The underlying idea behind this construction is that AI investments on their own might not be enough to actually build and implement AI systems, which then might have an effect on employment. Every AI start-up is likely to depend on qualified technicians and professionals, knowing how to construct, implement and use AI. Hence, one could argue that AI VC investments can only be truly effective and lead to actual AI implementation if there was already a qualified workforce, which is why this instrument is called effective intensity of AI. Obviously, AI firms can hire qualified personnel as they need it. However, there is an

argument that the implementation of AI systems might be facilitated with an already existing number of qualified workers at the start of the period of observation, as these workers might bring industry-specific knowledge with them. This facilitated implementation might affect how much is invested compared to the cross-country industry trend, either attracting even more investments or decreasing the need for large increases in investment. The share of ICT workers in 2011 can be assumed exogenous with the change in AI VC investments from 2012 to 2019 if there are no anticipation effects or other pre-trends affecting both. However, the identifying assumption for this IV hinges on the uncorrelatedness of the interaction itself, that is that the effective intensity of AI does not affect employment through any other channel.

As the exclusion restriction of the instrument could potentially be violated, given that it is interacted with the industry AI VC investments, I also propose a second instrument which might be more plausibly exogenous: the change in effective performance of AI. It is again an interaction with the share of ICT workers in each country-industry cell in 2011 but this time it is interacted with the change in the average training compute of notable ML systems. These are usually the most powerful and famous AI algorithms available like Google's AlphaGo from 2016 (Sevilla et al., 2022). Even though, commercial AI systems from start-ups are probably less powerful, this can serve as a benchmark of how the technological frontier in this field evolved over time. Again, I argue that pure computational power alone is not enough to actually lead to commercial AI systems and the success-

ful implementation of them, necessitating an existent skilled workforce and that differing changes in effective performance of AI affect the increase in investments. The identifying assumption is that the interaction is exogenous, which is facilitated if the number of workers in ICT occupations in 2011 is exogenous and the evolution of the training compute of milestone AI models is only driven by exogenous technological progress. Table 2 reports the first stage results for both instruments. Perhaps surprisingly, the coefficients for both instruments are negative. A possible explanation may be that higher effective intensity or effective performance of AI, through its facilitated implementation of AI, might decrease the need for AI VC investments slightly[4].

[4]Note that the extremely small magnitudes of the coefficients for the second instrument are mainly due to the comparatively large increases of average training compute in this time period. It increased from around 160 petaFLOPS to over 30 million petaFLOPs.

Table 2—First Stage Results for Both Instruments

	Δ AI Intensity	
	(1)	(2)
Panel A: IV Effective Intensity of AI		
Δ Effective AI Intensity	-0.011***	-0.019***
	(0.0002)	(0.0005)
Panel B: IV Effective Performance of AI		
Δ Effective AI Performance	-0.00000037***	-0.00000059***
	(0.0000001)	(0.0000002)
Observations	378	238
FE	Yes	Yes
Demographics	Yes	Yes
ICT Investments		Yes
Chinese Imports		Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note:ΔAI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included in designated specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons emplyoed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares.

As can be seen in table 3, both first-stage relationships are not particularly strong, given a Kleibergen-Paap F-statitsic, which is heteroscedasticity and cluster robust (Kleibergen and Paap, 2006), of around 14. The table also summarizes the second stage results using 2SLS for both instruments. Columns 1 and 2 show results for employment, columns 3 and 4 for hours worked, and columns 5 and 6 for hourly wages. Using the first instrument, effective intensity of AI, one can see that the 2SLS results for the change in log employment and log hours worked are strongly negative, although the magnitude decreases by around a half as more controls are included. In both cases they are significant at the 0.1 percent level and both coefficients are more negative compared with the OLS results in table 1. The decrease in the magnitude can be explained by ICT investments. AI and ICT investments are positively correlated. Should ICT investments have a negative effect on employment, they might bias the AI coefficient downwards. Perhaps surprisingly, the estimated effect on hourly wages seems to be positive and also highly significant, suggesting differential magnitudes of displacement and productivity effects for employment and wages. Again, the effect is much larger than OLS suggested.

Table 3—2SLS Results for Employment, Hours Worked and Average Wages

	Employment		Hours Worked		Hourly Wages	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: IV Effective Intensity of AI						
ΔAI Intensity	-0.80***	-0.47***	-0.89***	-0.53***	0.47**	0.45**
	(0.12)	(0.06)	(0.12)	(0.10)	(0.12)	(0.15)
F-Statistic	32.1	14.6	32.1	14.6	32.1	14.6
AR Wald CS Lower	-1.14; -0.29	-0.57	-1.23	-0.86	-0.53; 0.32	0.26
AR Wald CS Upper	-0.65; 0.17	-0.08	-0.74	-0.41	-0.02; 0.80	0.92
Panel B: IV Effective Performance of AI						
ΔAI Intensity	-0.83***	-0.48***	-0.95***	-0.58***	0.51**	0.53*
	(0.14)	(0.05)	(0.13)	(0.11)	(0.15)	(0.20)
F-Statistic	13.3	13.9	13.3	13.9	13.3	13.9
AR Wald CS Lower	-1.41; -0.56	-0.65	-1.36	-0.95	0.35	0.29
AR Wald CS Upper	-0.68; 0.08	-0.41	-0.09	-0.44	1.05	1.17
Observations	378	238	378	238	378	238
FE	Yes	Yes	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes	Yes	Yes
ICT Investments		Yes		Yes		Yes
Chinese Imports		Yes		Yes		Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: ΔAI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included in designated specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons emplyoed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares. The F-statistic is the Kleibergen-Paap F-statistic. The Anderson-Rubin (AR) confidence sets are constructed by inverting the weak instrument robust Anderson-Rubin test at the significance level of 5 percent. If there are two values for the upper and lower bound, then the confidence set has a hole, for instance in column 1 of panel A it would be [-1.14, -0.65]∪[-0.29, 0.17].

The effects of worker displacement seem to outweigh potential increases in employment due to productivity gains. As the effect on employment is very close to that of hours worked, mainly the extensive margin seems to be affected. Wages, however, seem to rise, suggesting productivity gains for those not laid off. The second instrument produces very similar coefficients compared to the first one, although effects for all outcome variables seem to be slightly stronger. Cautiously moving past mere correlations, one can interpret the 2SLS results as suggesting a negative and significant effect on the growth of employment and hours worked as well as a positive and significant effect wage growth. If taken at face value, the final coefficients imply that an increase of 100 dollars in AI VC investments per worker over the period of 2012-2019 is associated with an approximate decrease of employment growth by 4.7 (4.8) percentage points, a decrease of 5.3 (5.8) percentage points for hours worked, and an increase of 4.5 (5.3) percentage points for hourly wages in a country industry cell on average[5]. Given that the mean increase of AI VC investments over this time period was around 38 dollars per worker, the overall effect would not have been very sizable yet.

[5]The effects in the table are for an increase of 1000 dollars per worker in AI VC investments. Due to the linearity of the model, the corresponding effect for an increase of 100 dollars, which is more realistic, is the coefficient divided by 10. Note that the interpretation as percentage points is an approximation. The exact effect is given by $e^{\frac{\beta}{10}} - 1$.

D. Other Effects

AI can also have effects on other parts of the production function, especially capital and labour productivity. Table 4 reports both OLS and 2SLS results for these two different outcome variables[6].

Column 1 and 2 summarise the results for the all-assets capital stock in each country-industry cell. The OLS coefficient is negative, whereas the 2SLS results are positive, although neither is statistically significant, suggesting no detectable effect on the capital stock. The next two columns summarise the findings for labour productivity. Both instruments find mainly positive results that increase in magnitude with the introduction of the controls. In the final 2SLS specifications, the effect is somewhat significant. Hence, there seem to be positive effects of AI VC investments for labour productivity but none for capital, which corroborates the theoretical setup in section II, assuming that automation technologies are introduced in the production process when they enhance productivity.

[6]The corresponding first stage results can be found in the appendix.

Table 4—OLS and 2SLS Results for Capital and Labour Productivity

	Capital Stock		Labour Productivity	
	(1)	(2)	(3)	(4)
Panel A: OLS				
AI Intensity	0.045	-0.045	0.071	-0.008
	(0.083)	(0.026)	(0.072)	(0.020)
Panel B: IV Effective Intensity of AI				
AI Intensity	0.19	0.20	0.03	0.42**
	(.)	(0.15)	(0.11)	(0.14)
F-Statistic	45.3	14.6	32.1	14.6
AR Wald CS Lower	-4.28	-0.02	-0.20	0.26
AR Wald CS Upper	0.52	0.62	0.20	0.89
Panel C: IV Effective Performance of AI				
AI Intensity	0.24	0.23	0.04	0.47*
	(0.17)	(0.22)	(0.15)	(0.16)
F-Statistic	12.0	15.4	13.3	13.9
AR Wald CS Lower	-1.97	-0.17	-0.41	0.23
AR Wald CS Upper	1.39	0.77	0.26	0.92
N	336	226	378	238
FE	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
ICT Investments		Yes		Yes
Chinese Imports		Yes		Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Δ AI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included in designated specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons emplyoed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares. The F-statistic is the Kleibergen-Paap F-statistic. The AR Confidence sets are constructed by inverting the weak instrument robust AR test at the significance level of 5 percent.

E. Robustness

As can be seen in the descriptive analysis of the OECD.AI data, real AI investments per worker are very concentrated across industries and countries. It is similar for the log change in AI VC investments, underlying the main measure of this analysis and one of the instruments.

Table 5—IV Results excluding Outliers

	(1) Employment	(2) Hours	(3) Wages	(4) Capital	(5) Productivity
Panel A: IV Effective Intensity of AI					
ΔAI Intensity	-2.188***	-2.643***	2.110**	0.290	1.773*
	(0.345)	(0.497)	(0.517)	(0.559)	(0.724)
F-statistic	21.2	21.2	21.2	21.3	21.2
Panel B: IV Effective Performance of AI					
ΔAI Intensity	-2.024***	-2.615***	2.306**	0.485	1.850
	(0.452)	(0.570)	(0.620)	(0.962)	(0.947)
F-statistic	35.1	35.1	35.1	29.9	35.1
Observations	224	224	224	212	224

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Country and industry fixed effects are included in all specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons employed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares. The F-statistic is the Kleibergen-Paap F-statistic. Regressions are weighted by 2011 within-country employment shares. Observations with a change in AI intensity of more than 191 dollars per worker, corresponding to the 95th percentile, are omitted.

Hence, I reexamine my results for all outcome variables, dropping observations with changes in AI investments above the 95th percentile to see whether the results are mainly driven by a few outliers. Table 5 reports the outcome. Surprisingly, I do not only report negative and significant results for employment and hours again but the effect increases substantially compared to my main analysis. The findings for hourly wages and labour productivity are also substantially bigger but barely significant for labour productivity.

An additional concern can be the use of the asymptotically valid clustered and robust standard errors in a setting with relatively few clusters[7]. Furthermore, Young (2022) argues that under non-iid errors, statistical inference can suffer and the bias of the IV can increase. He demonstrates this by using, inter alia, the bootstrap to show that a substantial share of published IV papers reject the null of their main hypotheses too often. As I have only a relatively small number of clusters, I also recreate the hypothesis tests for my 2SLS results using the wild bootstrap proposed by Finlay and Magnusson (2014). Table 6 reports the results. The effect of the small sample can be seen as with the bootstrap, I am only able to reject the null for the effect on employment growth using effective performance as an IV.

[7] 27 country clusters and 14 industry clusters (13 if Chinese exports are included).

Table 6—2SLS Results with Bootstrapped P-Values

	Employment	Hours	Wages	Capital	Productivity
Panel A: IV Effective Intensity of AI					
ΔAI Intensity	-0.47	-0.53	0.45	0.20	0.42
P-value	0.109	0.203	0.132	0.307	0.150
Panel B: IV Effective Performance of AI					
ΔAI Intensity	-0.48*	-0.58	0.53	0.23	0.47
P-value	0.045	0.118	0.107	0.305	0.079
Observations	238	238	238	226	238

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: ΔAI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons employed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). The wild bootstrap-t is used with small sample adjustment and 999 draws.

Being in the fortunate position to have two instruments available, I also check whether including both of them at the same time alters the results. As can be seen in table 7, there are still negative and significant effects for employment, hours worked, and the share of women, while I report positive and significant effects for wages and labour productivity. The magnitudes only change marginally. What this exercise shows is that the existence and sign of the effects of AI seem to be mostly robust to outliers and between the choice of instruments, even though the magnitudes are less so. However, inference using the bootstrap suggests that most results might not be

significant and should be viewed with utmost caution.

Table 7—IV Results with Both Instruments Included

	(1) Employment	(2) Hours	(3) Wages	(4) Capital	(5) Productivity
ΔAI Intensity	-0.46***	-0.51***	0.42*	0.19	0.40*
	(0.07)	(0.10)	(0.14)	(0.13)	(0.14)
J-statistic	0.12	1.64	1.32	0.10	0.50
	[0.734]	[0.201]	[0.250]	[0.751]	[0.481]
Effective F-statistic	11.1	11.1	11.1	11.2	11.1
Observations	238	238	238	226	238

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Country and industry fixed effects are included in all specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons emplyoed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. Regressions are weighted by 2011 within-country employment shares. J-statistic refers to the Hansen J-statistic of the overidentification test (Hansen, 1982). P-values of the J-test are in square brackets.

V. Discussion

Having presented the results of my analysis, it is imperative to discuss the limitations. As already mentioned in section IV.C, the OLS coefficients are likely to suffer from multiple sources of endogeneity like omitted confounders, reverse causality, and measurement issues. To tackle these problems, I use the two instrumental variables presented above. The internal validity of an instrumental variable approach with homogeneous treatment

effects, hinges on two key assumptions: instrument relevance and instrument validity. Relevance needs the instrument to have a high enough correlation with the endogenous regressor. A typical way to test it and rule out a weak instrument is using the F-statistic of the first stage (Wooldrige, 2016). I use the Kleibergen-Paap F-statistic as it is robust to heteroscedasticity and clustering (Kleibergen and Paap, 2006). Since the first instrument's F-statistic usually hovers around 14, I feel not confident enough to completely rule out a weak instrument problem. Therefore, I report the weak instrument robust Anderson-Rubin confidence sets, obtained by inverting the Anderson-Rubin test (Anderson and Rubin, 1949). They seem to corroborate my findings as can be seen in tables 3 and 4.

The second concern is generally not testable. Instrument validity requires that the instrument should have no partial effect on the outcome variable and be uncorrelated with the omitted variables (Wooldrige, 2016). In the case of effective intensity of AI, this means, as already explored, that the interaction of the number of ICT workers in each country-industry cell and the AI investments of an industry should have no partial effect on emplyoment, hours worked, and any other outcome variable explored. This is not impossible, especially if the share of ICT workers in 2011, before the period of observation, is unrelated with anything else that could have impacted the outcome variables. The second instrument, effective performance of AI, is based on the argument that pure computational power of AI systems alone is not enough to implement AI systems and skilled workers are needed. The validity of this instrument also hinges on the uncorrelatedness of the interac-

tion. As ML training compute of the most powerful AI models in the world is probably mainly due to technological constraints, which are unlikely to affect the labour market through other channels than AI implementation, this instrument could plausibly satisfy validity, especially if the share of ICT workers in 2011 is indeed exogenous. Therefore, I regard the validity of this instrument with more confidence.

However, there are multiple threats to validity. Countries could experience different shocks that also affect AI, especially other technology shocks like the introduction of industrial robots. I do not include them as controls since they are mainly reported for a few manufacturing-adjacent industries, thus severely limiting my sample size. Furthermore, any of the included controls could introduce inconsistency as one contaminated regressor contaminates the entire analysis. Finally, the AI VC data suffer from measurement errors as some VC deals had to be estimated. The portion of estimated investments depends largely on the country. Furthermore, there are still a sizeable portion of deals that do not have reported or estimated values for the amount of investment, reaching 56 percent of all deals in Slovenia. On top of these issues comes the fact that some deals do only enter the data set after a delay, so especially for the newest years, there might be more missing investments (Tricot, 2021). Finally, I only have data on AI startups but AI is also developed by publicly traded firms, so I am not able to capture their effects. Thus, a violation of the IV assumptions cannot be ruled out, especially given the close relationship with the endogenous regressor for the first instrument. A way to analyse the exclusion restriction is an overidenti-

fication test (Hansen, 1982). As I have two instruments I can carry out the Sargan-Hansen overidentifaction test in a specification using both instruments. As can be seen in table 6, I can never reject the null of having two jointly valid instruments, strengthening my results, especially since the estimated effects are similar to the ones using only one instrument. However, high p-values of this test are probable, even when instruments are invalid (Kiviet and Kripfganz, 2021). Hence, a failure to reject the null, although encouraging, is not a proof of joint validity.

As regards external validity, the evidence at the country-industry level seems to contradict some other findings in the AI literature. Using data on AI related job vacancies in the US, Acemoglu et al. (2022) find that firms exposed to artificial intelligence reduce hiring in non-AI-related positions. However, they do not find significant effects for employment or wages at the industry level. Their firm-level results can be consistent with the findings in this analysis, although the industry-level results are not. As Felten, Raj and Seamans (2019) and Fossen and Sorgner (2019), I find positive effects on wage growth.

VI. Conclusion

Are we moving towards the technological singularity with accelerating speed, losing more and more relevance as machines destroy jobs? Or is this a repetition of past arguments and grievances towards automating technologies?

Artificial intelligence is certainly an impressive technology and expected

to have significant effects in the future. However, this analysis suggests that these effects might already exist. Using an instrumental variable approach, I analyse the effect of AI VC investments on different labour market outcomes. I find negative effects for growth in employment and hours worked, as well as positive ones for growth in hourly wages and productivity. Estimating aggregate effects and including other developed countries, this work adds to a literature dominated by firm- and individual-level evidence from the US. Unlike in the existing literature, I find negative employment effects but report positive effects on wage growth as do Felten, Raj and Seamans (2019) and Fossen and Sorgner (2019). Given the limitations of the methodology used, these results can only be regarded as a tentative step towards a causal impact. Hence, it is of upmost import to collect more comprehensive and complete data for the implementation of AI. This is not an easy task given the immaterial nature of AI. With the rise of more general AI applications like ChatGPT, which can be used for a multitude of different tasks, spillovers abound and it becomes less possible to consign the effects to a specific industry. New approaches are thus needed. Given the potential epochal transformations expected by AI, its effects at the micro as well as the macro level need to be better understood. At least for now, it is still up to humans to actually conduct this kind of research, even if writing it down might not be anymore.

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Derivation of the Specification

Acemoglu and Restrepo (2020) find the following equation for the effect of robots in local labour markets:

$$d \ln L_c = \psi \sum_{i \in I} l_{ci} \frac{d\theta_{ci}}{1-\theta_{ci}} \frac{\gamma_L}{\gamma_M} \tag{A1}$$

where the ζ from section II is $\psi \frac{\gamma_L}{\gamma_M}$ and again a function of model parameters. l_{ci} is the labour share of industry i in commuting zone c. Note that this expression can be rewritten as the weighted average of the industry effects:

$$d \ln L_c = \sum_{i \in I} l_{ci} \psi \frac{\gamma_L}{\gamma_M} \frac{d\theta_{ci}}{1-\theta_{ci}}$$

$$d \ln L_c = \sum_{i \in I} \frac{L_{ci}}{L_c} \zeta \frac{d\theta_{ci}}{1-\theta_{ci}}$$

$$d \ln L_c = \frac{\sum_{i \in I} \zeta \frac{d\theta_{ci}}{1-\theta_{ci}} L_{ci}}{L_c}$$

$$d \ln L_c = \frac{\sum_{i \in I} d \ln L_{ci} L_{ci}}{L_c}$$

The expression inside the sum thus contains the industry-wide effect of automation and is given by:

$$d \ln L_{ci} = \zeta \frac{d\theta_{ci}}{1-\theta_{ci}} \tag{A2}$$

The same argument applies for wages.

Matching of Industries

Table B1—Industry Matching

NACE Rev.2	OECD.AI
A: Agriculture, Forestry and Fishing	R: Agriculture
B-D-E: Mining, utilities and water	H: Energy, raw materials and utilities
C: Manufacturing	J: Robots, sensors, IT hardware; K: consumer products; L: Mobility
F: Construction	Q: Construction and air conditioning
G-H: Wholesale and retail trade; transportation and storage	I: Logistics, wholesale and retail
I: Accommodation and food services	G: Travel, leisure, hospitality; N: food and beverages
J: Information and Communication	B: IT infrastructure; D: Media, social platforms, marketing; M: Digital security
K: Financial and insurance activities	C: Finance and Insurance
L: Real estate activities	T: Real estate
M-N: Professional, scientific, technical, administrative and support activities	E: Business processes, support; O: Environmental services
O: Public administration and defence	P: Government, security and defence
P: Education	A: Education and Training
Q: Human health and social work	F: Healthcare, drugs and biotechnology
R-S: Arts, entertainment, recreation and other services	S: Consumer services

Note: The one letter industry code for the OECD.AI data is my own coding scheme.

Additional Tables

Table C1—First Stage Results for Table 4 Regressions

	Δ AI Intensity			
	(1)	(2)	(3)	(4)
Panel A: IV Effective Intensity of AI				
Δ Effective AI Intensity	-0.012***	-0.019***	-0.011***	-0.019***
	(0.0002)	(0.0005)	(0.0002)	(0.0005)
Panel B: IV Effective Performance of AI				
Δ Effective AI Performance	-0.00000039***	-0.00000060***	-0.00000037***	-0.00000059***
	(0.0000001)	(0.0000002)	(0.0000001)	(0.0000002)
Observations	336	226	378	238
FE	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
ICT Investments		Yes		Yes
Chinese Imports		Yes		Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note:ΔAI Intensity refers to the change in AI investments divided by the initial number of workers in the country-industry cell. Country and industry fixed effects are included in designated specifications. Demographics include the share of employed persons above 50, the share of women, and the share of workers having attended university. ICT investments and Chinese imports are transformed as AI Intensity. Effective Intensity in AI is the interaction between the industry AI VC investments except for the country observed with the share of persons emplyoed in ICT occupations in 2011. Effective Performance of AI is the same interaction with the with the average difference in training compute. Both instruments are then transformed as the change in AI Intensity. Standard errors are robust, two-way clustered by country and industry and displayed in parentheses. There are 27 country clusters and 14 industry clusters (13 when Chinese Imports are included). Regressions are weighted by 2011 within-country employment shares.